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DIGIHAZ

Digital Transformation for Natural Hazards

High-Resolution Imagery for Disaster Mapping

Drone-Based Damage Assessment Applications

Module 04 - Topic #4

Instructor: Dr. Giorgos SFIKAS (Assistant Professor, University of West Attica)

DIGIHAZ is implemented by:

Universitat d'Alacant
Universidad de Alicante
University of Alicante
Cluster Lead
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The University of West Attica
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Learning Objectives

By the end of this lecture you will be able to:

01 Understand the need for automatic change detection, using 2D or 3D techniques.

02 Understand the way photogrammetric products such as DEMs and 3D point clouds are used for disaster mapping and damage assessment.

03 Learn about debris estimation and automatic volume change analysis using UAV sensor-based data.

Topics Covered

- Pre- and Post-Disaster Image Comparison
- Detecting Collapsed Infrastructure and Height Reduction
- Volume Change Analysis for Landslides
- Debris Estimation for Cleanup Planning

Estimated Duration
45 minutes lecture + 45 min exercise

Pre- and Post-Disaster Image Comparison

- Change detection

Pre- and Post-Disaster Image Comparison

- Change detection
- 2D & 3D change analysis

Pre- and Post-Disaster Image Comparison

- Change detection
- 2D & 3D change analysis
- 2D -- reliance on pixel-2-pixel comparison of orthomosaics

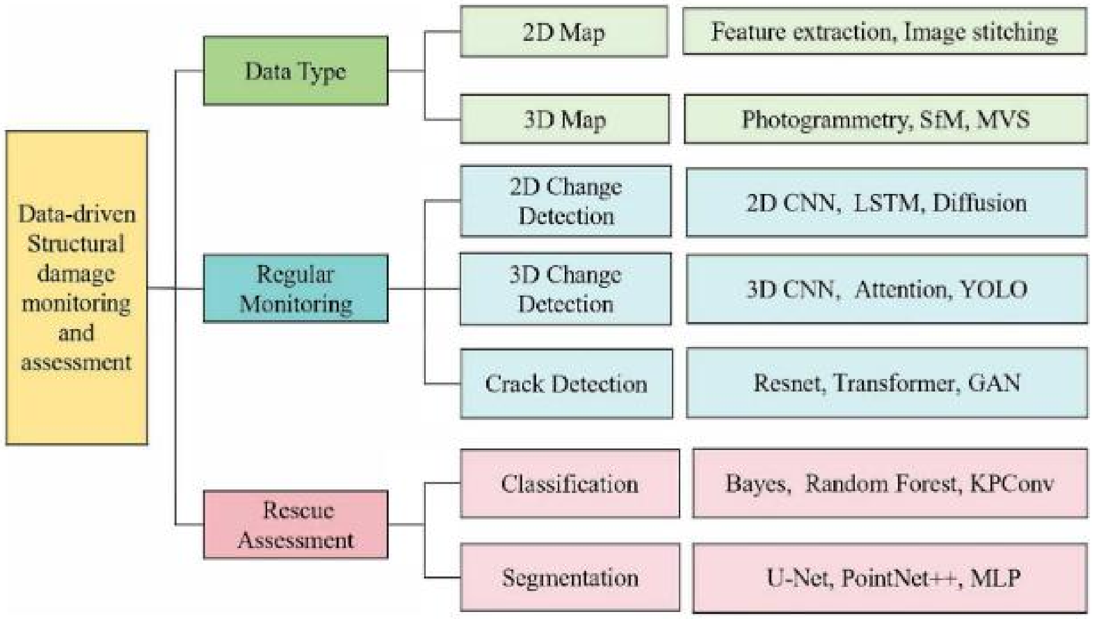
Pre- and Post-Disaster Image Comparison

- Change detection
- 2D & 3D change analysis
- 2D -- reliance on pixel-2-pixel comparison of orthomosaics
- Advanced ML can bring in more complex cues

Pre- and Post-Disaster Image Comparison

- Change detection
- 2D & 3D change analysis
- 2D -- reliance on pixel-2-pixel comparison of orthomosaics
- Advanced ML can bring in more complex cues
- e.g. Convolutional Neural Networks

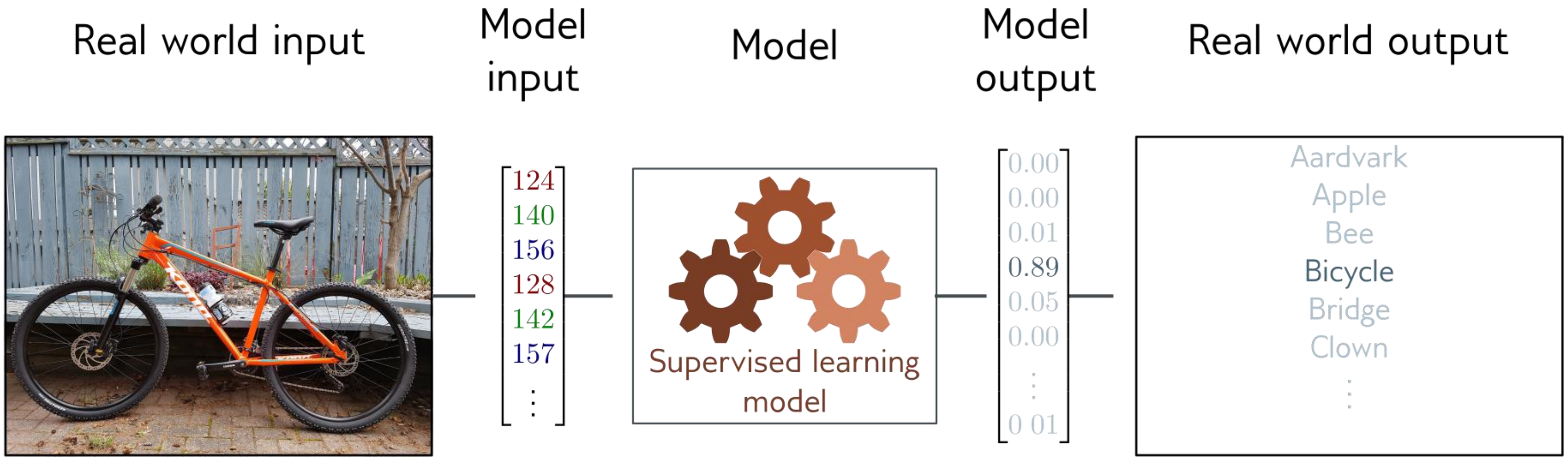
Drone Imagery Processing and Photogrammetric Workflows



The architecture of data-driven structural damage monitoring and rescue assessment methods. The type of data and the target tasks influence the selection of the damage assessment methods

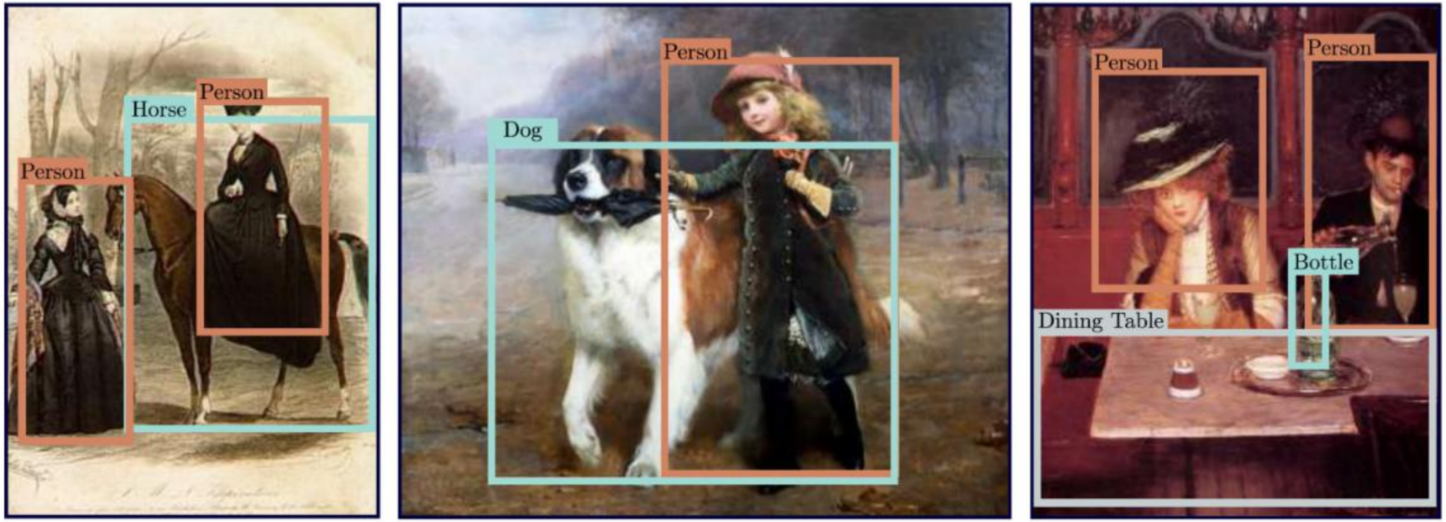
Pattern Recognition & Machine Learning methods

Classification



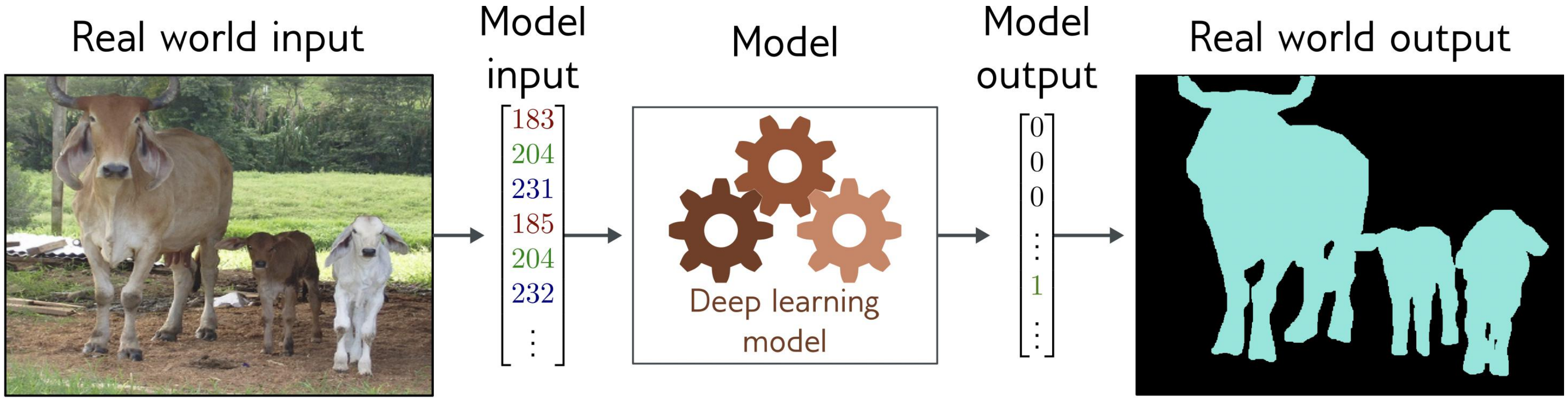
Pattern Recognition & Machine Learning methods

Object detection

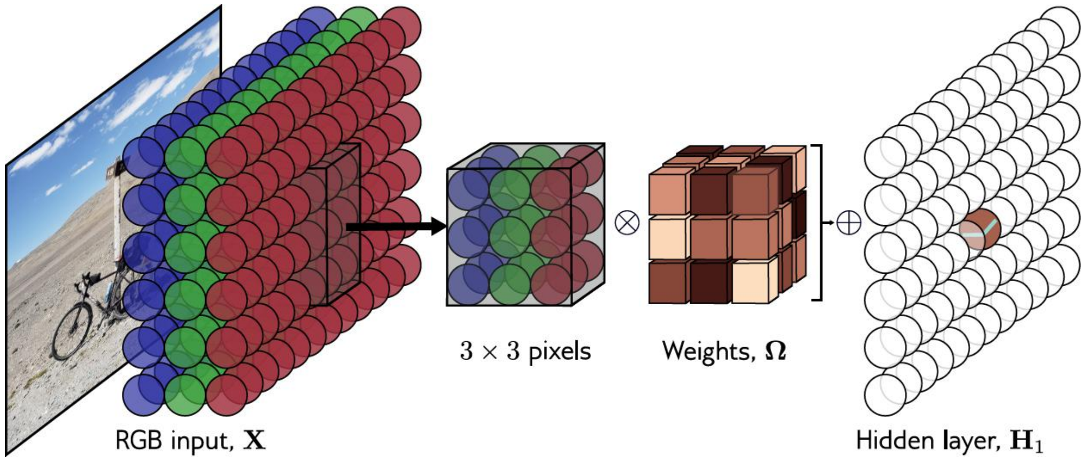


Drone Imagery Processing and Photogrammetric Workflows

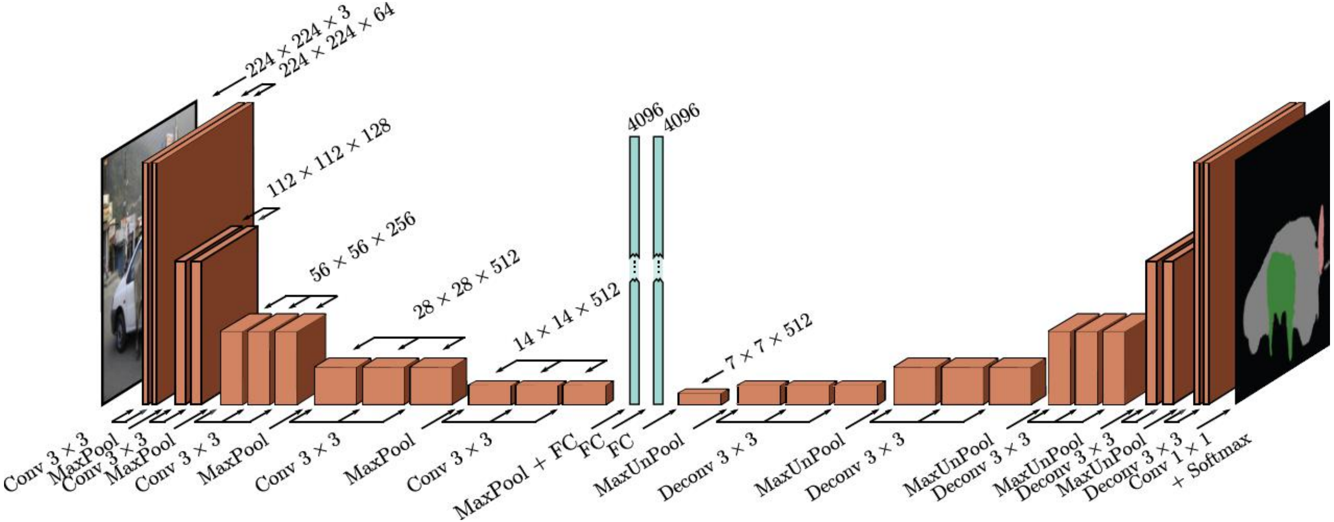
Pattern Recognition & Machine Learning methods
Segmentation



Pattern Recognition & Machine Learning methods
Convolutional Neural Networks



Pattern Recognition & Machine Learning methods
An encoder-decoder network for Semantic Segmentation



Girisha, S., et al. "Semantic segmentation of UAV aerial videos using convolutional neural networks." 2019 IEEE International Conference on Artificial Intelligence and Knowledge Engineering (AIKE). IEEE, 2019.

Fig. 1. Example images and labels from UAVid dataset. First row shows the images captured by UAV. Second row shows the corresponding ground truth labels. Third row shows the prediction results of MS-Dilation net + PRT + FSO model as in Table 1.

Lyu, Y., Vosselman, G., Xia, G. S., Yilmaz, A., & Yang, M. Y. (2020). UAVid: A semantic segmentation dataset for UAV imagery. ISPRS Journal of Photogrammetry and Remote Sensing, 165, 108-119.

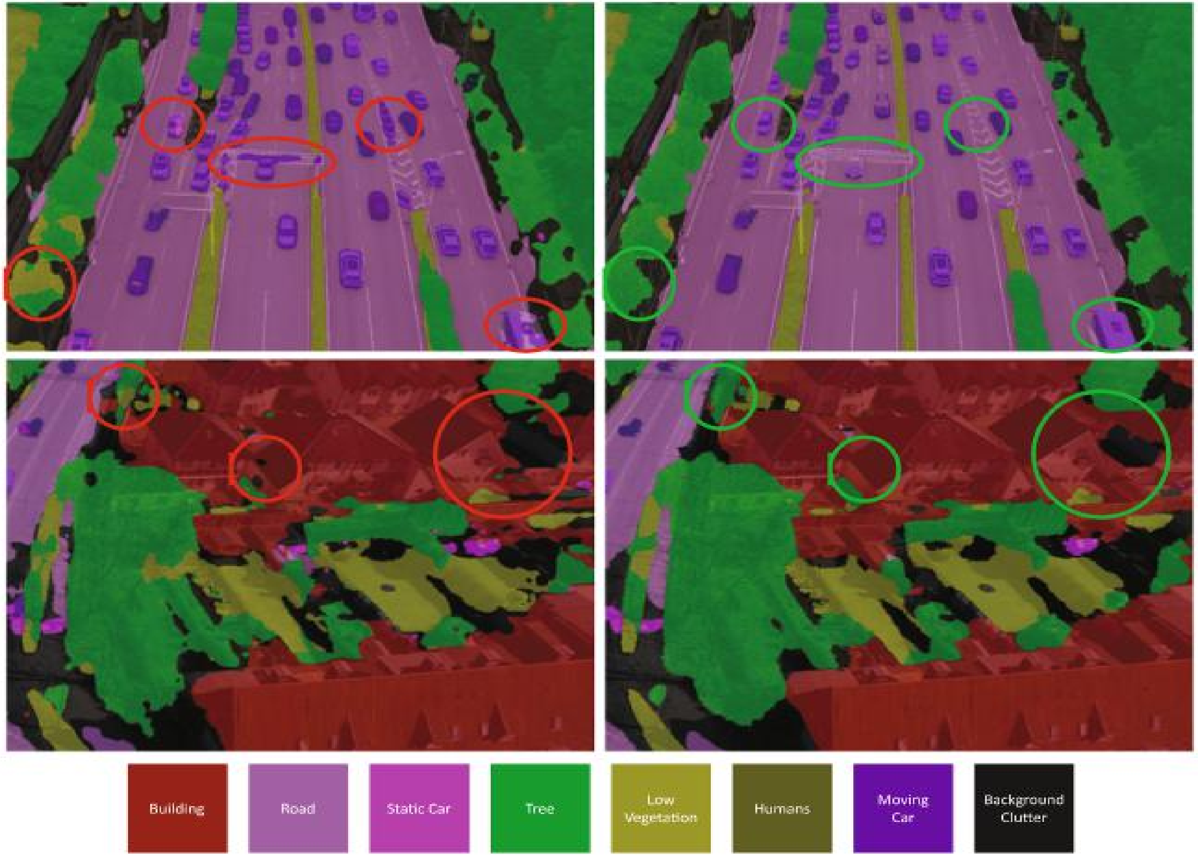
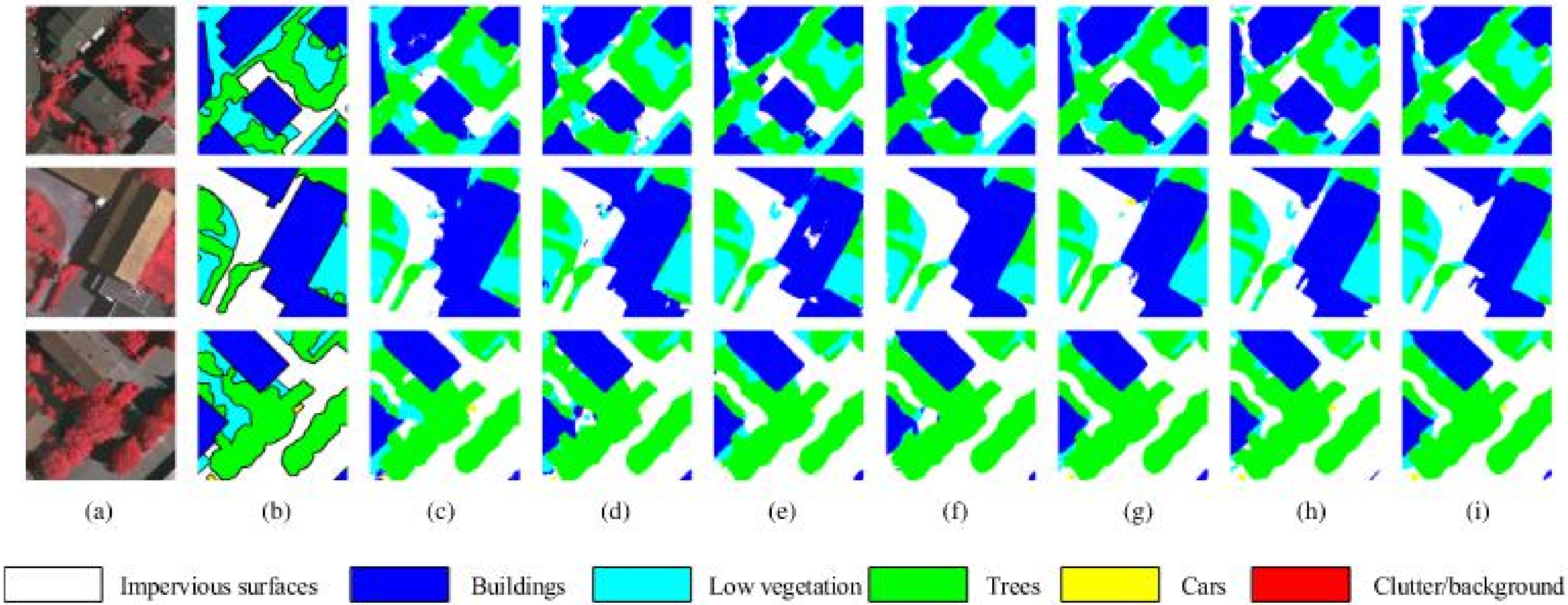


Fig. 12. Examples of spatial-temporal regularization for UAVid image semantic segmentation. The left column shows the prediction without FSO plus 3D CRF refinement. The right column shows the corresponding refined prediction with FSO plus 3D CRF refinement. The most obvious improvements are high-lighted with circles. The spatial-temporal regularization achieves a more coherent prediction for different objects.

Lyu, Y., Vosselman, G., Xia, G. S., Yilmaz, A., & Yang, M. Y. (2020). UAVid: A semantic segmentation dataset for UAV imagery. ISPRS Journal of Photogrammetry and Remote Sensing, 165, 108-119.



Liu, Siyu, et al. "Light-Weight Semantic Segmentation Network for UAV Remote Sensing Images." IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing 14 (2021): 8287-8296.

(b) SlimYOLOv3-SPP3-95

Figure 8. Visualized detection results of SlimYOLOv3-SPP3-95 and YOLOv3-SPP3 on a challenging frame captured by our drone.

Zhang, Pengyi, Yunxin Zhong, and Xiaoqiong Li. "SlimYOLOv3: Narrower, faster and better for real-time UAV applications." Proceedings of the IEEE/CVF International Conference on Computer Vision Workshops. 2019.

Pre- and Post-Disaster Image Comparison

- Limitations of 2D analysis

Pre- and Post-Disaster Image Comparison

- Limitations of 2D analysis
- 3D change detection as crucial for structural damage assessment

Pre- and Post-Disaster Image Comparison

- Limitations of 2D analysis
- 3D change detection as crucial for structural damage assessment
- Precise differences via 3D point cloud and DSM comparison

Pre- and Post-Disaster Image Comparison

- Limitations of 2D analysis
- 3D change detection as crucial for structural damage assessment
- Precise differences via 3D point cloud and DSM comparison
- Identification of changes not visible in a 2D analysis

Detecting Collapsed Infrastructure and Height Reduction

- Height reduction analysis

Detecting Collapsed Infrastructure and Height Reduction

- Height reduction analysis
- Subtraction of post-event DSM to pre-event DSM

Detecting Collapsed Infrastructure and Height Reduction

- Height reduction analysis
- Subtraction of post-event DSM to pre-event DSM
- Automated process vs manual inspection

Volume Change Analysis for Landslides

- Volume change critical as a cue for landslides

Volume Change Analysis for Landslides

- Volume change critical as a cue for landslides
- Comparing pre- and post- event elevation

Volume Change Analysis for Landslides

- Volume change critical as a cue for landslides
- Comparing pre- and post- event elevation
- Precision depends on sensor type -- LiDAR vs Structure from Motion

Debris Estimation for Cleanup Planning

- Total volume estimation as a logistical prerequisite

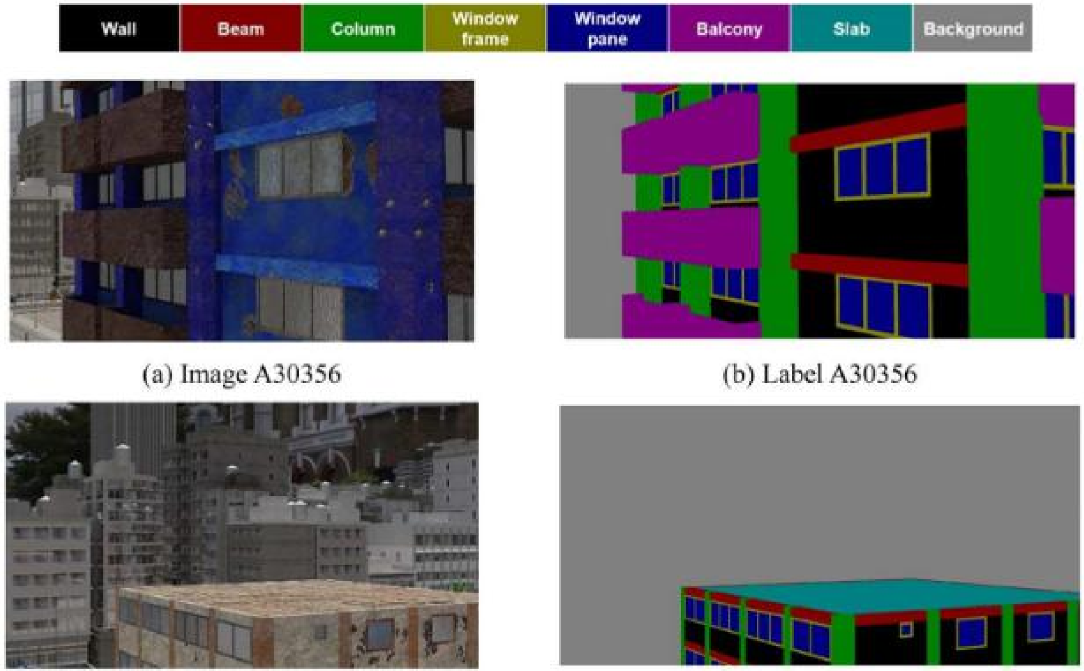
Debris Estimation for Cleanup Planning

- Total volume estimation as a logistical prerequisite
- Volume estimation via integration

Debris Estimation for Cleanup Planning

- Total volume estimation as a logistical prerequisite
- Volume estimation via integration
- Deep learning & UAV photogrammetry

Drone Imagery Processing and Photogrammetric Workflows



(a) Image A30356

(b) Label A30356

(c) Image B30040

(d) Label B30040

Damage assessment based on segmentation methods.

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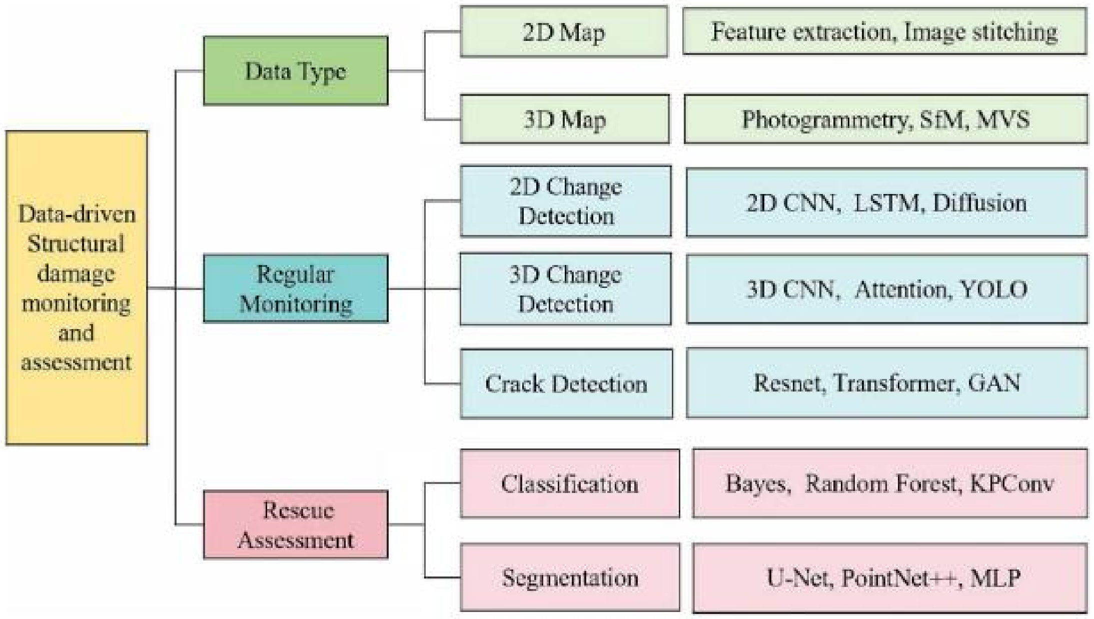
Drone Imagery Processing and Photogrammetric Workflows



The architecture of data-driven structural damage monitoring and rescue assessment methods. The type of data and the target tasks influence the selection of the damage assessment methods

Key terms & definitions

Key Terms & Definitions	
3D Change Detection	The process of comparing pre- and post-disaster elevation models or point clouds to compute precise volumetric and geometric differences, allowing analysts to identify vertical collapse or structural deformations that are not visible in traditional top-down 2D imagery.
Height Reduction Analysis	An automated, algorithmic method used to detect completely or partially collapsed infrastructure by mathematically subtracting a post-event Digital Surface Model (DSM) from pre-event baseline elevation data.
Volume Change Analysis	An analytical method used for mass movement geohazards, such as landslides, to map and quantify displaced material by evaluating the vertical differences between pre- and post-event elevation models.

Key terms & definitions

Key Terms & Definitions	
Zones of Depletion and Accumulation	In landslide volumetric analysis, the zone of depletion refers to the area where material has fallen away (resulting in a negative volume change), while the zone of accumulation is the area where the displaced material has finally settled (resulting in a positive volume change).
Zonal Statistics	A spatial analysis operation within Geographic Information Systems (GIS) used to calculate specific metrics—such as the total volume of a debris field or the maximum height of a building—by aggregating and evaluating raster pixel values within a defined geometric boundary or polygon.

Discussion questions

1. How are change detection methods useful in pre- and post- disaster comparisons? What are some differences between 2D and 3D methods, in terms of either mechanism or capability?
2. In landslides, how are zones of accumulation and zones of depletion differentiated?
3. How are debris pile volumes calculated in an urban disaster setting?

Sneak peak at your exercise

Example

We have seen how UAV imagery is crucial in geospatial monitoring and post-hoc estimation. Orthophotos & DEMs are important products in this respect. In this exercise, you will be tasked to create an orthophotography given a Digital Elevation Model and UAV imagery data.

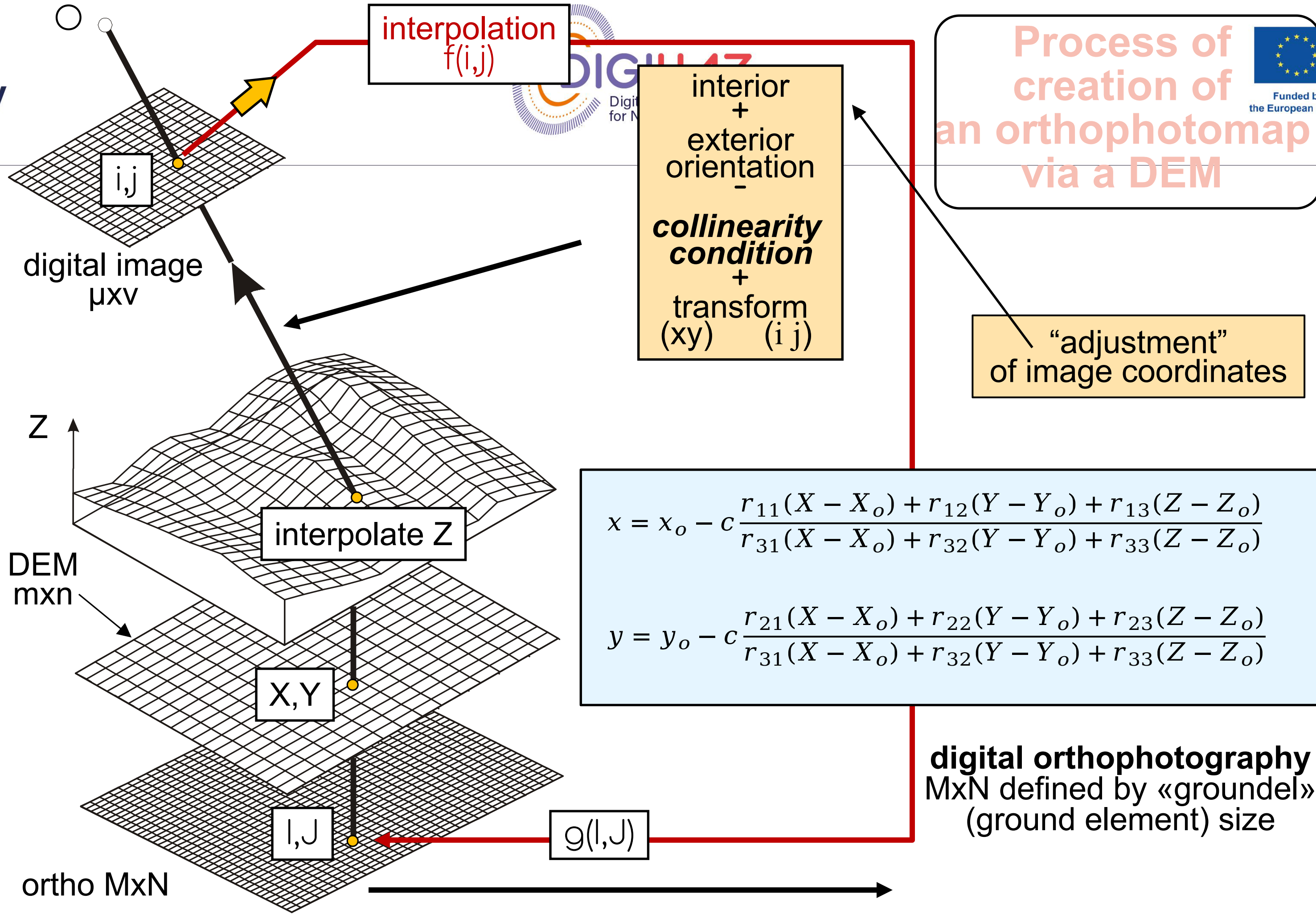
See also:

GitHub repo / Notebook link

Exercise steps:

1. Load the Digital Elevation Model.
2. Load the UAV imagery data.
3. Use the collinearity condition to map X,Y,Z from the DEM to the imaging data.
4. Display the orthophotomap.

Process of creation of an orthophotomap via a DEM



$$x = x_o - c\frac{r_{11}(X - X_o) + r_{12}(Y - Y_o) + r_{13}(Z - Z_o)}{r_{31}(X - X_o) + r_{32}(Y - Y_o) + r_{33}(Z - Z_o)}$$

$$y = y_o - c\frac{r_{21}(X - X_o) + r_{22}(Y - Y_o) + r_{23}(Z - Z_o)}{r_{31}(X - X_o) + r_{32}(Y - Y_o) + r_{33}(Z - Z_o)}$$

digital orthophotography
MxN defined by «groundel»
(ground element) size

Sneak peak at your exercise

Example

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See also:

GitHub repo / Notebook link

```
# Drone image size
width = 8192          # Image size (in pixels)
height = 5460

# Interior orientation
c = 35.9235           # Focal length
xo = -0.1130          # Principal point
yo = -0.0680
d_im = 0.0044         # Pixel dimension (in mm)

# Exterior orientation (DJI_0106.jpg)
Xo = 302412.9194
Yo = 4134628.6581
Zo = 655.9317
omega = 0.2158
phi = 0.2202
kappa = -42.62469
```

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Sneak peak at your exercise

Example

We have seen how UAV imagery is crucial in geospatial monitoring and post-hoc estimation. Orthophotos & DEMs are important products in this respect. In this exercise, you will be tasked to create an orthophotography given a Digital Elevation Model and UAV imagery data.

See also:

GitHub repo / Notebook link

```python
#
# Load 'DJI_0106.jpg' as a python array.
# TODO
# im = .....

# Load the elevation model 'dsm_2.5cm.tif' as a python array.
# TODO
# dsm = .....
# dsm = dsm[:3001, :2001] ## Use this to obtain a smaller version of the data,
to speed up your experiment.

# Check dimensions of loaded DSM.
# TODO
# height_dsm, width_dsm = ....
# print(dsm.shape)
```

Python

Sneak peak at your exercise

Example

We have seen how UAV imagery is crucial in geospatial monitoring and post-hoc estimation. Orthophotos & DEMs are important products in this respect. In this exercise, you will be tasked to create an orthophotography given a Digital Elevation Model and UAV imagery data.

```python
# TODO: Convert degrees into radians for Euler angles ω, φ, κ.
#omega = ..
#phi =
#kappa =

# TODO: Compute corresponding rotation matrices to each Euler angle,
# and compute rotation matrix for the "complete" rotation by multiplying
# the 3 matrices (one for each Euler angle)
# R = ....
```

Python

See also:

GitHub repo / Notebook link

Sneak peak at your exercise

Example

We have seen how UAV imagery is crucial in geospatial monitoring and post-hoc estimation. Orthophotos & DEMs are important products in this respect. In this exercise, you will be tasked to create an orthophotography given a Digital Elevation Model and UAV imagery data.

See also:

GitHub repo / Notebook link

$$R_\omega = \begin{bmatrix} 1 & 0 & 0 \\ 0 & cos\omega & sin\omega \\ 0 & -sin\omega & cos\omega \end{bmatrix}$$

$$R_\phi = \begin{bmatrix} cos\phi & 0 & -sin\phi \\ 0 & 1 & 0 \\ sin\phi & 0 & cos\phi \end{bmatrix}$$

$$R_\kappa = \begin{bmatrix} cos\kappa & sin\kappa & 0 \\ -sin\kappa & cos\kappa & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

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Example

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$$R_\omega = \begin{bmatrix} 1 & 0 & 0 \\ 0 & cos\omega & sin\omega \\ 0 & -sin\omega & cos\omega \end{bmatrix}$$

$$R_\phi = \begin{bmatrix} cos\phi & 0 & -sin\phi \\ 0 & 1 & 0 \\ sin\phi & 0 & cos\phi \end{bmatrix}$$

$$R_\kappa = \begin{bmatrix} cos\kappa & sin\kappa & 0 \\ -sin\kappa & cos\kappa & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$R = R_\kappa R_\phi R_\omega$$

Sneak peak at your exercise

Example

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See also:

GitHub repo / Notebook link

```
for i in tqdm(range(height_ortho // d_groundel)):
    for j in range(width_ortho // d_groundel):
        # Correspondence of DSM coordinates to 3D space coordinates X,Y,Z.
        X = orX_ortho + (j-1)*d_ortho*d_groundel
        Y = orY_ortho - (i-1)*d_ortho*d_groundel
        # elevation interpolation
        # TODO Pick up "Z" using the value of point (i,j) in matrix 'dsm'.
        # Z = ....

        # Project X,Y,Z (DSM-based coordinates) to x,y (projected image plane)
        # Use the collinearity condition
        #U = ...
        #V = ...
        #W = ...
        x = xo - c*U/W
        y = yo - c*V/W
```

Sneak peak at your exercise

Example

We have seen how UAV imagery is crucial in geospatial monitoring and post-hoc estimation. Orthophotos & DEMs are important products in this respect. In this exercise, you will be tasked to create an orthophotography given a Digital Elevation Model and UAV imagery data.

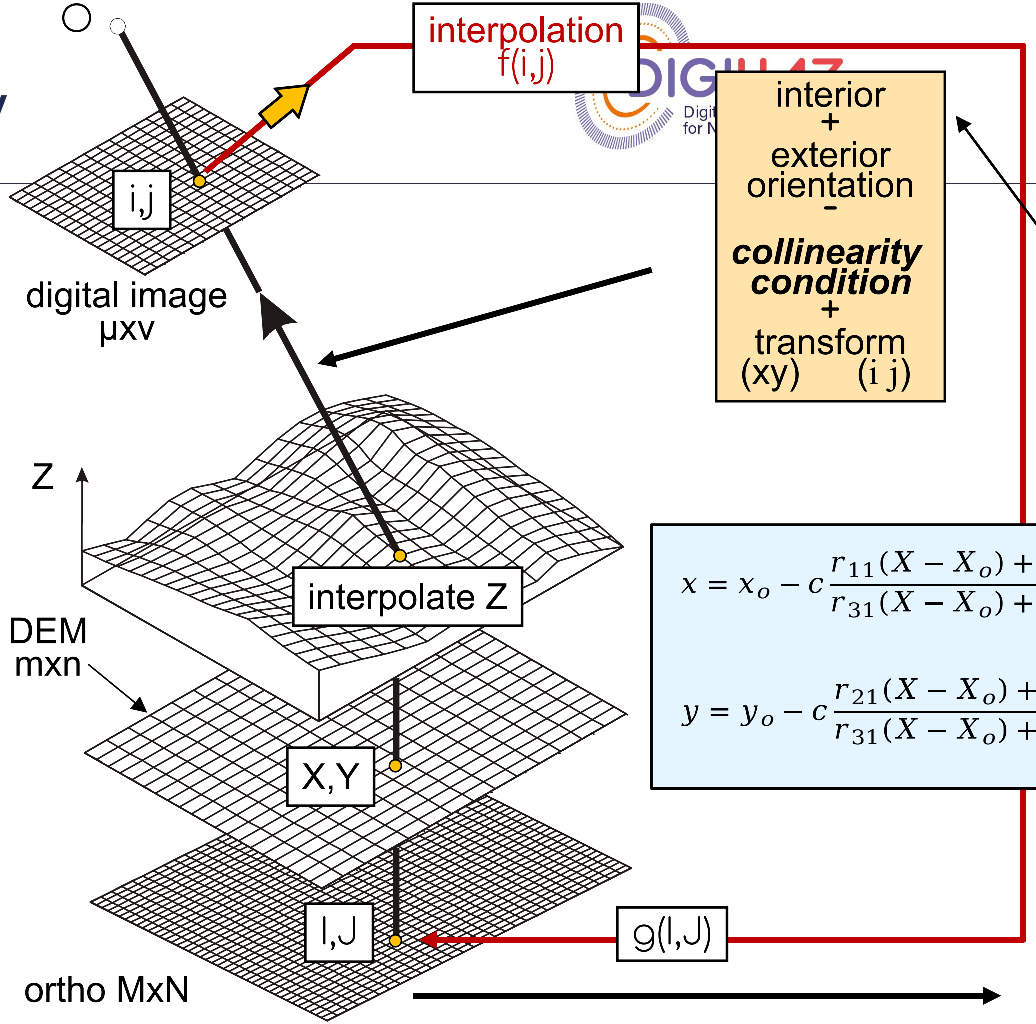
See also:

GitHub repo / Notebook link

```python
# Visualize and save the orthoimage.
# TODO
```

Python

Process of creation of an orthophotomap via a DEM



"adjustment"
of image coordinates

$$x = x_o - c\frac{r_{11}(X - X_o) + r_{12}(Y - Y_o) + r_{13}(Z - Z_o)}{r_{31}(X - X_o) + r_{32}(Y - Y_o) + r_{33}(Z - Z_o)}$$

$$y = y_o - c\frac{r_{21}(X - X_o) + r_{22}(Y - Y_o) + r_{23}(Z - Z_o)}{r_{31}(X - X_o) + r_{32}(Y - Y_o) + r_{33}(Z - Z_o)}$$

digital orthophotography
MxN defined by «groundel»
(ground element) size

Thank you!

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